

Patent #3 Non-Provisional — Amendment Addendum

New Specification Text for Continuation Anchoring

Date: June 28, 2026

Purpose: Inject the following sections into the Patent #3 non-provisional specification to secure priority date coverage for Patents #4 and #5

Instructions to Counsel

The following text blocks must be added to the specification of the non-provisional application. They do not modify existing claims — they add **embodiment descriptions** and **architectural context** that enable future Continuation Applications under 35 U.S.C. §120.

Each section is marked with its purpose and the patent it anchors.

Section A: Trust Assertion Definition

Insert after: The existing VEO definition section **Anchors:** All future patents (foundational framing)

Trust Assertion. A Verifiable Entropy Object (VEO) constitutes a cryptographically verifiable trust assertion regarding a non-deterministic computation. The trust assertion states that a specified output was generated using identified entropy sources, at a quantified quality level, with a specific cryptographic signature, under a declared policy,

achieving a defined trust level. The trust assertion is the primary output of the system; the entropy is the input material from which trust is established.

Section B: Verifiable Decision Object (VDO) Embodiment

Insert after: The Decision Receipt / Decision Attestation section **Anchors:** Patent #4 (Trust Coordination / Decision Verification)

Verifiable Decision Object. In one embodiment, the system supports a Verifiable Decision Object (VDO) — a structured, cryptographically signed digital object that encapsulates the complete context of a non-deterministic computational decision. A VDO contains: the identity of the deciding entity (which may be a human operator, an autonomous software agent, or a multi-agent system); the input parameters available at decision time; the applicable policy constraints; one or more referenced Verifiable Entropy Object identifiers that provided the non-deterministic input; the decision outcome; and independently verifiable attestation evidence including a cryptographic signature and optional anchoring to an immutable verification substrate.

A VDO enables post-hoc reconstruction and audit of autonomous decisions without requiring trust in the deciding entity. A third party holding a VDO can independently verify that the stated decision was made using the stated entropy, under the stated policy, by the stated entity, without access to any private keys or internal system state.

Example VDO structure:

```
json { "vdo_version": "1.0", "decision_id": "dec_a1b2c3d4", "decision_type":  
"tool_selection", "deciding_entity": { "entity_type": "autonomous_agent", "entity_id":  
"agent_007", "model_version": "gpt-4-2026-06" }, "inputs": { "available_tools": ["Tool A",  
"Tool B", "Tool C"], "context_hash": "sha256:abcdef..." }, "policy": { "policy_id":  
"fair_selection_v1", "constraints": ["equal_probability", "no_repeat_within_10"] },  
"entropy": { "veo_id": "veo_x9y8z7", "ecs_score": 847, "ecs_grade": "AA" }, "outcome": {  
"selected": "Tool B", "selection_index": 1, "confidence": 0.81 }, "trust_score": 720,  
"attestation": { "signature": "0x...", "signer_address": "0x...", "timestamp": "2026-06-  
28T05:30:00Z", "anchor": null } }
```

Application domains for Verifiable Decision Objects include but are not limited to: autonomous AI agent tool selection and routing, multi-agent arbitration and resource allocation, autonomous vehicle navigation decisions, robotic task assignment, algorithmic financial trading, healthcare workflow automation, enterprise process automation, and regulatory compliance workflows.

Section C: Trust Non-Amplification Invariant

Insert after: Section B (VDO Embodiment) **Anchors:** Patent #4 and Patent #5 (foundational mathematical property)

Trust Non-Amplification Invariant. The system enforces a Trust Non-Amplification Invariant: a derivative trust object's computed trust score cannot mathematically exceed the trust score of the root Verifiable Entropy Object(s) contributing to it. Trust is conserved or degraded through computational processing — it is never amplified.

For example, a Verifiable Decision Object consuming a VEO with an Entropy Confidence Score of 720 cannot itself claim a trust score exceeding 720. If a VDO references multiple VEOs, its maximum achievable trust score is bounded by the minimum ECS score among the referenced VEOs.

This invariant ensures that: 1. The trust system is mathematically self-correcting — no computational process can manufacture trust from inferior inputs 2. Trust propagation through chains of delegation is bounded and auditable 3. System-wide trust quality can be monitored by tracking trust scores at the leaves and verifying the invariant holds at every node 4. Artificial trust inflation through re-wrapping or re-signing is prevented by construction

The Trust Non-Amplification Invariant distinguishes the present system from standard execution logging and distributed tracing systems, which record events without enforcing conservation laws on trust quality.

Section D: Five-Layer Trust Stack Architecture

Insert after: Section C (Trust Non-Amplification Invariant) **Anchors:** Entire patent family portfolio coherence

Five-Layer Trust Stack. The present invention operates as the third layer of a five-layer trust architecture designed to provide end-to-end computational trust for non-deterministic systems:

Layer 1 — Trust Source (Randomness Layer). Generating verifiable entropy from multiple independent sources, including public randomness beacons, blockchain-derived entropy, quantum random number generators, and hardware random number generators. Each source produces independently verifiable entropy with provenance metadata.

Layer 2 — Trust Integrity (Integrity Layer). Preventing manipulation and preview of generated entropy through sequential computational proofs, including Verifiable Delay Functions, that create a physical temporal barrier requiring sequential hardware computation. This layer ensures that no party — including the generator — can preview or manipulate entropy values before they are committed.

Layer 3 — Trust Evidence (Trust Artifact Layer). Packaging entropy into Verifiable Entropy Objects with multi-dimensional quality scoring, cryptographic signing, optional anchoring to an immutable verification substrate, and multi-level verification. This layer — the subject of the present application — transforms disposable entropy into portable, auditable trust assertions.

Layer 4 — Trust Coordination (Decision Layer). Linking trust artifacts to specific computational decisions through Verifiable Decision Objects, preserving trust across delegation boundaries when computation passes between autonomous agents, microservices, organizations, or computing environments. This layer ensures that end-to-end verifiable trust survives delegation, enforcing the Trust Non-Amplification Invariant across all derivative trust objects.

Layer 5 — Trust Runtime (Governance Layer). Continuous, system-wide trust health monitoring, decentralized compliance verification, and governance enforcement across a distributed network of trust-producing and trust-consuming entities. The Trust Runtime monitors trust assertion quality metrics including entropy confidence score drift, source concentration risk, delegation chain depth, anchoring latency, and trust score distribution. The Trust Runtime enforces the Trust Non-Amplification Invariant network-wide and generates governance alerts when trust health indicators fall below configurable thresholds.

The architecture exhibits the following properties: each layer depends on the layers below it; trust flows upward through the stack and is conserved via the Trust Non-Amplification Invariant; verification can occur at any layer independently; and the stack is composable — applications may use Layers 1 through 3 without requiring Layers 4 and 5.

Section E: Trust Health Monitoring Embodiment

Insert after: Section D (Five-Layer Trust Stack) **Anchors:** Patent #5 (Trust Runtime / Continuous Trust Health)

Trust Health. In one embodiment, the system further comprises a Trust Runtime that continuously monitors trust health across a distributed network of VEO producers and consumers. Trust Health is defined as the continuous, quantitative measurement of computational trust quality across a system of interacting components that produce, consume, and propagate trust assertions.

Trust Health is analogous to established system health metrics: - Battery Health measures charge capacity degradation over charge cycles - Cluster Health measures distributed system coordination quality across nodes - Trust Health measures trust assertion quality across producers, consumers, and delegation chains over time

The Trust Runtime monitors the following trust health indicators:

1. **ECS Drift** — tracking how Entropy Confidence Scores change across a fleet of VEO-producing systems over time, detecting systematic quality degradation
2. **Source Concentration Risk** — detecting when an excessive proportion of trust assertions depend on an insufficient number of independent entropy sources, creating systemic vulnerability
3. **Delegation Chain Depth** — measuring how many delegation steps separate a trust assertion from its root entropy, flagging chains where trust degradation accumulates
4. **Anchoring Latency** — monitoring the elapsed time between VEO generation and immutable substrate anchoring, detecting delays that increase the vulnerability window
5. **Trust Score Distribution** — observing system-wide trust score patterns for statistical anomalies indicating manipulation, source failure, or systematic bias

6. Trust Non-Amplification Compliance — continuously verifying that no derivative trust objects in the network violate the Trust Non-Amplification Invariant

The Trust Runtime enables continuous compliance verification without requiring re-audit of individual trust assertions, providing system-wide governance over computational trust quality. This monitoring axis is fundamentally distinct from security monitoring (which tracks threats and vulnerabilities), performance monitoring (which tracks latency and throughput), and availability monitoring (which tracks uptime). Trust Health monitors the quality and integrity of trust assertions themselves.

Checklist Before Filing

- ☐ All five sections (A–E) incorporated into specification
- ☐ No specific implementation names in independent claims (Polygon, SHA-256, secp256k1, EIP-191, Ethereum)
- ☐ Independent claims use generalized terminology per Term Mapping table in Instructions v4
- ☐ VDO JSON structure present in specification
- ☐ Trust Non-Amplification Invariant defined
- ☐ Five-Layer Trust Stack described with all five layers
- ☐ Trust Health embodiment paragraph included
- ☐ 101 defense grounded in hardware computational constraints (VDF as physical temporal barrier)
- ☐ Cross-references to related applications updated